

Chapter 3

Procurement Processes

Learning objectives

- Procurement and Logistics Execution.
- Material Management Master Data and Material Master Data.
- Procure-To-Pay Process.
- Materials Management in SAP S/4HANA (using Global Bike).

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1. Procurement and Logistics Execution

Solution Map of SAP ERP

- The SAP ERP solution map

Functionality
of SAP ERP

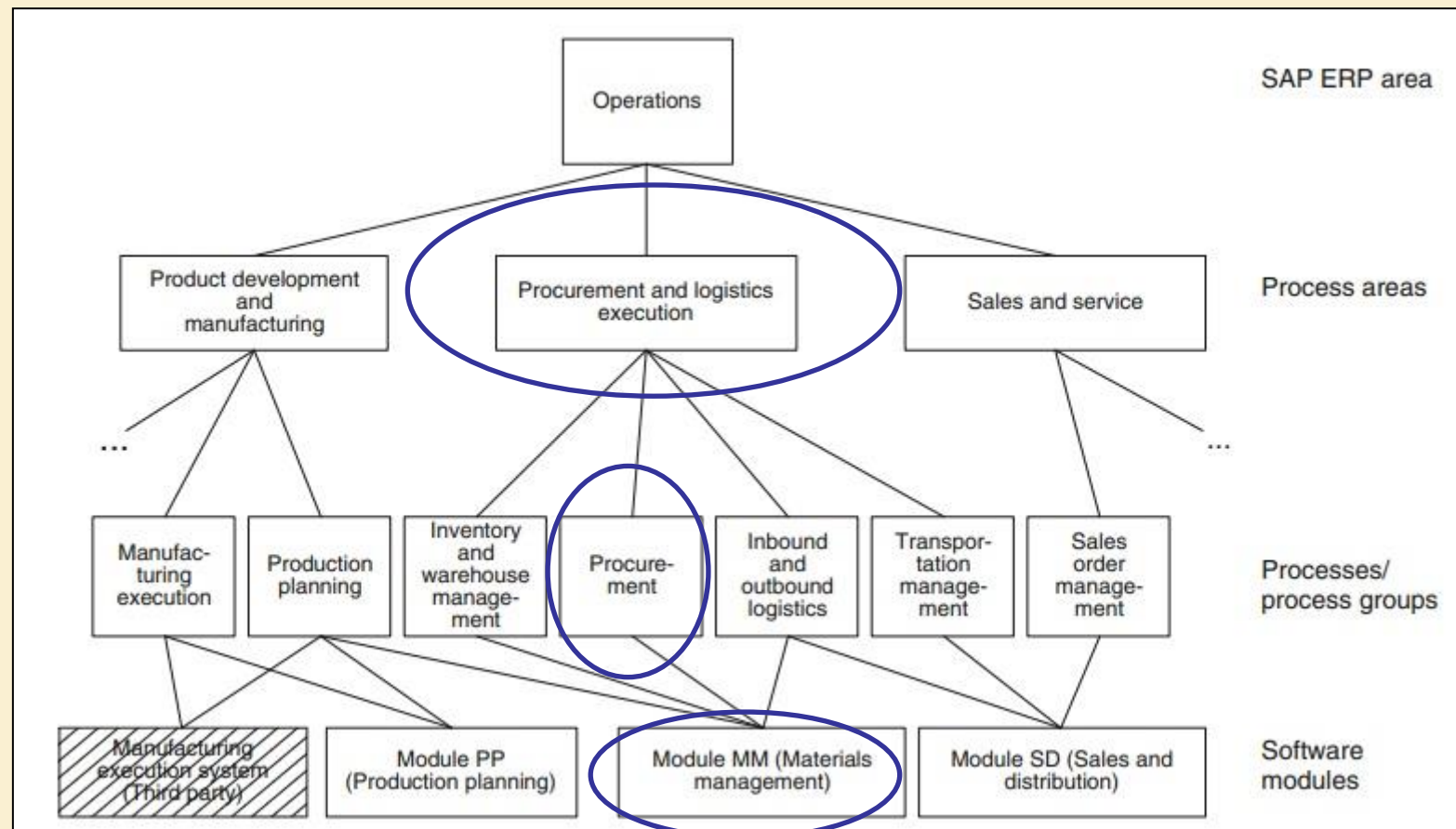
	End-user service delivery										Shared service delivery	SAP NetWeaver
Analytics	Financial analytics			Operations analytics			Workforce analytics					
Financials	Financial supply chain management		Treasury		Financial accounting		Management accounting		Corporate governance			
Human capital management	Talent management			Workforce process management			Workforce deployment					
Procurement and logistics execution	Procurement		Inventory and warehouse management		Inbound and outbound logistics			Transportation management				
Product development and manufacturing	Production planning		Manufacturing execution		Product development			Life-cycle data management				
Sales and service	Sales order management			Aftermarket sales and service			Professional-service delivery					
Corporate services	Real estate management	Enterprise asset management	Project and portfolio management	Travel management	Environment, health, and safety compliance management		Quality management	Global trade services				

Procurement and Logistics Execution

- The processes and tasks of procurement and logistics execution are primarily supported by functions of the modules MM (materials management) and SD (sales and distribution). This area includes
 - ❖ Procurement.
 - ❖ Inventory and warehouse management.
 - ❖ Inbound and outbound logistics.
 - ❖ Transportation management.

Procurement and Logistics Execution (Cont.)

- Relationship between solution map and SAP ERP modules



Procurement and Logistics Execution - Inventory and Warehouse Management

- The inventory and warehouse management part offers support for a variety of tasks, including:
 - ❖ Inbound processing (i.e., follow-on activities for a purchase order after goods have been received).
 - ❖ Outbound processing (delivery processing and distribution, goods issue documents, proof of delivery).
 - ❖ Cross-docking (goods are brought directly from the receipt of the goods to goods issued without being stored).
 - ❖ Physical inventory (periodic, continuous, sampling, etc.)
 - ❖ Warehousing and storage.

Procurement and Logistics Execution - Inbound and Outbound Logistics

- Inbound logistics cover all steps in procurement after the goods have been received (goods receipt, transportation to warehouses/storage places according to the stock placement strategy, etc.).
- Outbound logistics include preparing and sending the goods to the recipients (goods removal, goods issues, shipping documents, etc.).
- When goods are imported or exported, the accompanying documents required by customs are created, and the duty costs are calculated.

Procurement and Logistics Execution - Transportation management

- Transportation management also uses Sales & Distribution functions for transportation planning (route determination, shipper selection, etc.), transportation execution (all activities connected with delivery), and freight costing.
- Freight costing includes calculating freight charges (considering conditions, payment arrangements, etc.) and freight cost settlement (including settlement for multiple carriers involved in a shipment).

Procurement and Logistics Execution - Procurement Process

- The procurement part of procurement and logistics execution comprises requisitioning (purchase requisitions), purchase order processing (calculating order requirements, planning, releasing, and monitoring the purchase order), receiving goods, handling returns, and the financial settlement (invoice verification and blocking and releasing invoices).

Procurement and Logistics Execution - Procurement Process (Cont.)

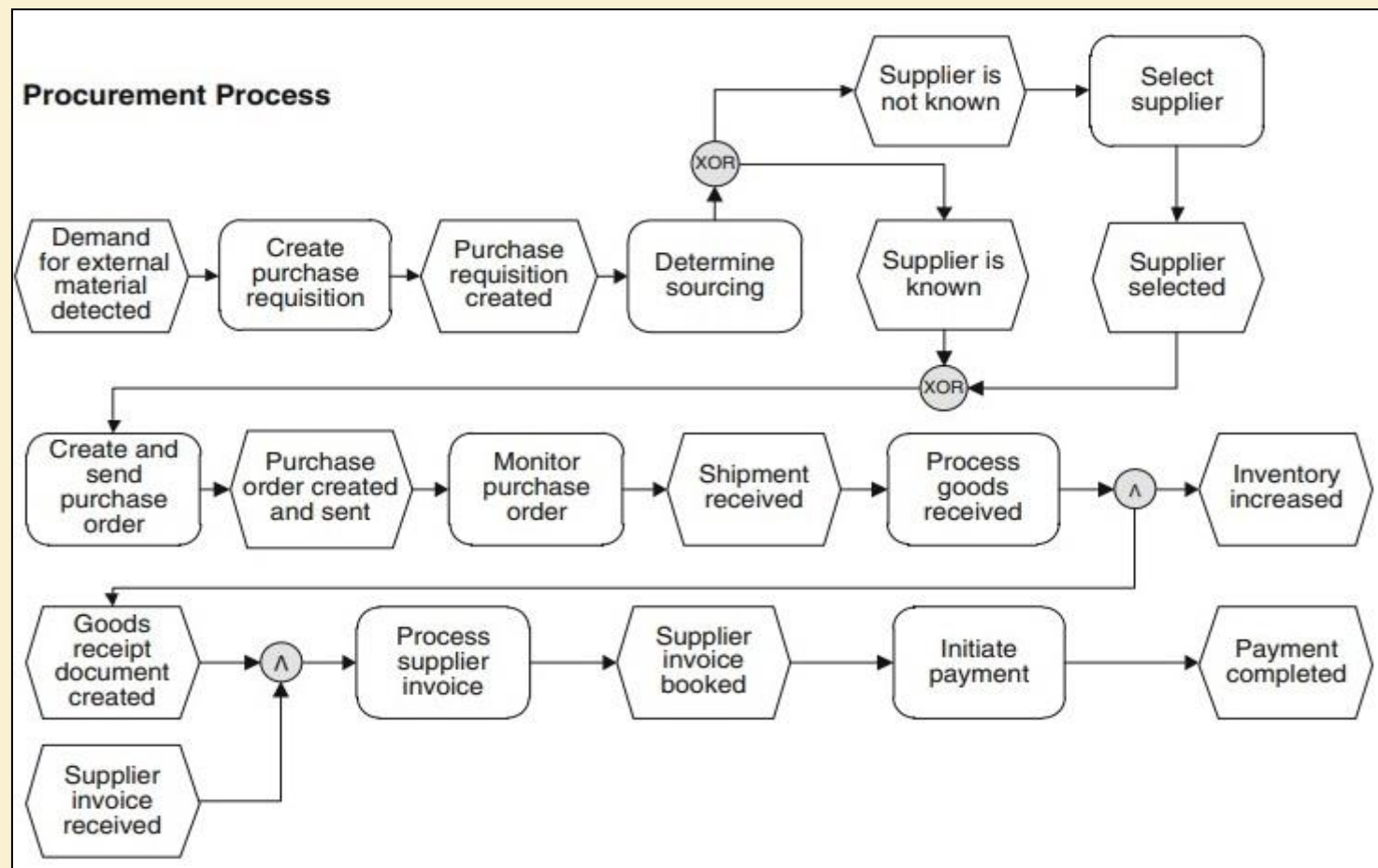
- The basic procurement process



- The above diagram indicates that the initial step in this process is to create a requisition, which is then converted to a purchase order and sent to a vendor.
- When the vendor receives the purchase order, it ships the materials, which the ordering party receives in the received materials step.
- The ordering party also receives an invoice from the vendor, and it then makes a payment to the vendor.

Procurement and Logistics Execution - Procurement Process (Cont.)

- The procurement process primarily applies functionality from materials management and accounting.



2. Material Management Master Data and Material Master Data

4500000000

Standard PO

Edit

Copy

Display Commitments

Status

Purchase Order Date

Net Value

Follow-On Documents

02.08.2021

5.250,00 USD

^

↺

General Information

Items

Limit Items

Delivery and Invoice

Supplier Contact Data

Notes

Output Management

Attachments >

▼

Basic Data

Purchase Order Type:
Standard PO (NB)

Supplier:
[Olympic Protective Gear \(101999\)](#)

Currency:
United States Dollar (USD)

Language Key:
English (EN)

Organization

Purchasing Group:
North America (N00)

Purchasing Organization:
Global Bike US (US00)

Company Code:
Global Bike Inc. (US00)

Further Information

Created by:
—

Purchase Order Date:
02.08.2021

Approval Status:
—

Approver:
—

Output Master Data

Vendor Master Data

- Vendor Master
 - ❖ Contains all the necessary information needed for business activities with an external supplier.
 - ❖ Used and maintained primarily by the Purchasing and Accounting Departments.
 - ❖ Every vendor **MUST** have a master record.

SAP Business Partner

Olympic Protective Gear
101000

Basic Data Roles Address Bank Accounts Payment Cards Identification Contacts Attachments

Business Partner Role	Valid From	Valid To
FI Vendor FLVN00	08/18/2021	12/31/9999
Vendor FLVN01	08/18/2021	12/31/9999

Address

Standard Address

Street: 2100 Summit Boulevard	Country: USA (US)	Post Office Postal Code: -
House Number: -	Region: Georgia (GA)	Post Box: -
Postal Code: 30319		
City: Atlanta		

Vendor Master Data (Cont.)

- Vendor Master
 - ❖ The supplier master record is created in SAP S/4HANA as a central business partner master record with different roles.
 - ❖ For suppliers, these are often the business partner roles:
 - FLVN00 (FI Vendor).
 - FLVN01 (Vendor).

SAP Business Partner

Olympic Protective Gear
101000

Basic Data ▾ Roles Address ▾ Bank Accounts Payment Cards Identification ▾ Contacts Attachments

Search

Business Partner Role	Valid From	Valid To
FI Vendor FLVN00	08/18/2021	12/31/9999
Vendor FLVN01	08/18/2021	12/31/9999

Address

Standard Address

Street: 2100 Summit Boulevard
Country: USA (US)
Post Office Postal Code: -
House Number: -
Region: Georgia (GA)
Post Box: -
Postal Code: 30319
City: Atlanta

Vendor Master Views

- Client Level
 - ❖ Address.
 - ❖ Vendor Number.
 - ❖ Preferred Communication.
- Company Code Data
 - ❖ Reconciliation Account.
 - ❖ Terms of Payment.
 - ❖ Bank Account.
- Purchase Org Data
 - ❖ Purchasing Currency.
 - ❖ Salesman's Name.
 - ❖ Vendor Partners.

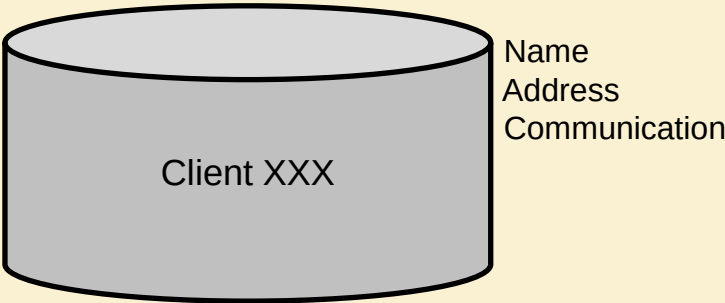
General Data

Company Code Data
Financial Accounting (FI)

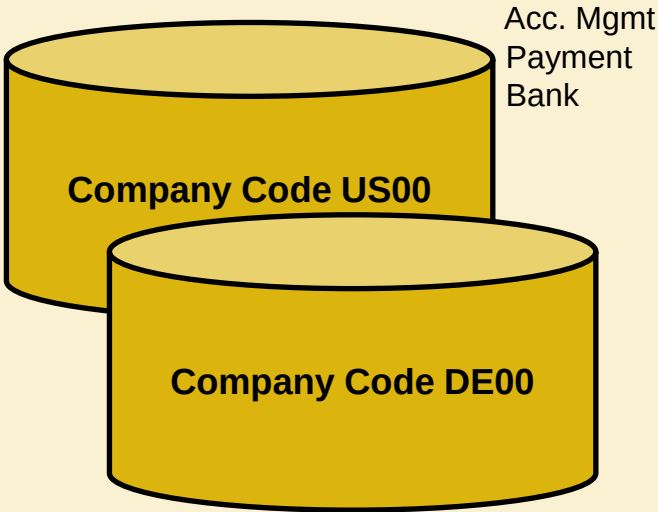
Purchasing Data
Materials Mgmt. (MM)

Vendor Master

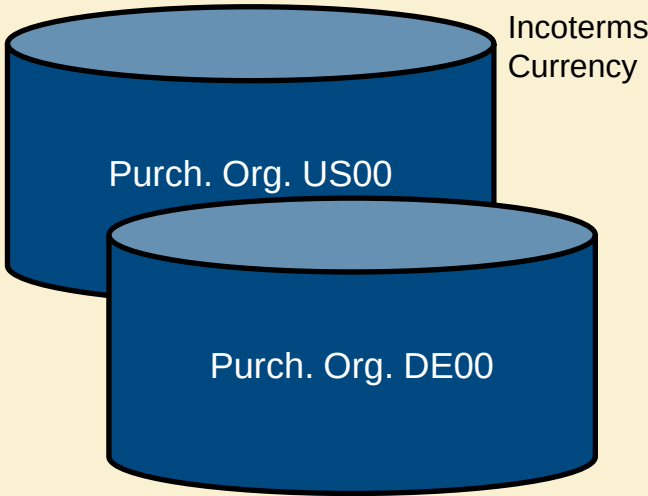
General Information relevant for the entire organization:



Company Code specific information:



Purch. Organization specific information:



Material Master Data

- Material Master
 - ❖ Contains all the information a company needs to manage about a material.
 - ❖ It is used by most components within the SAP system.
 - Sales and Distribution.
 - Materials Management.
 - Production.
 - Plant Maintenance.
 - Accounting/Controlling.
 - Quality Management.
 - ❖ Material master data is stored in functional segments called Views.

SAP Product

Deluxe Touring Bike (black)
DXTR1000

Edit Copy

Product Type: Finished Product (FERT) Base Unit of Measure: Each (EA) Revision Level:
Product Category: Product GTIN:
Product Group: Finished Bikes (BIKES) GTIN Category:

General Information Product Compliance Components Texts Sales Storage Warehouse Management

Basic Data

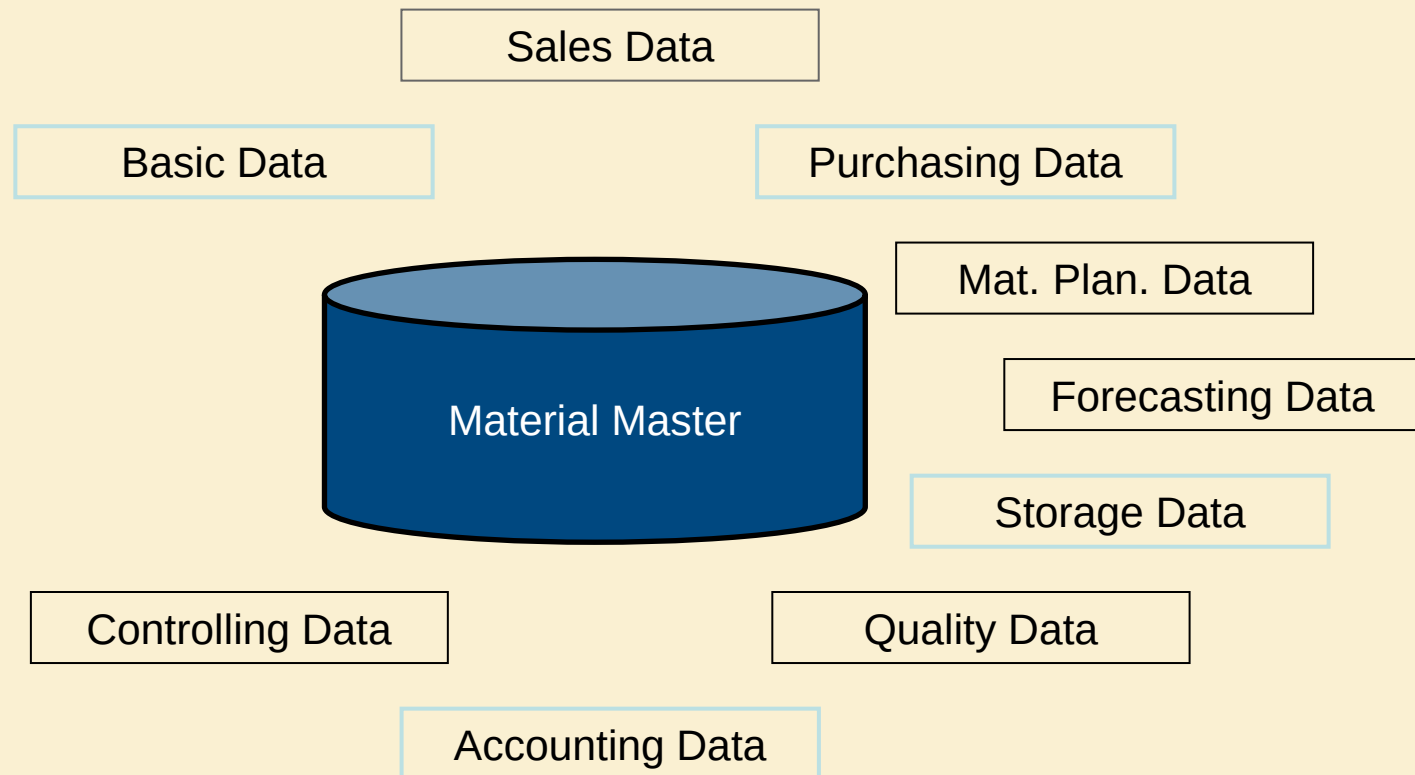
Division: Bicycles (BI) Created By: Chris Reich
Old Product Number: — Created On: 08/19/2021, 11:55:19
Batch Management Required: No Last Changed By: Chris Reich
Marked for Deletion: No Last Changed On: 08/23/2021, 12:36:09

Show More

Descriptions

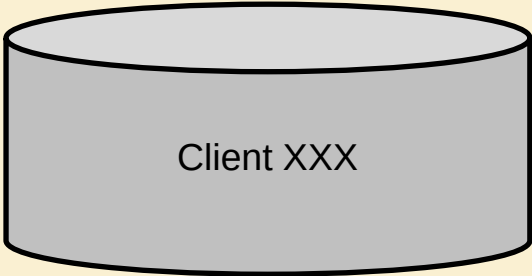
Language	Product
German DE	Deluxe Touring Bike (schwarz)
English EN	Deluxe Touring Bike (black)

Material Master Views



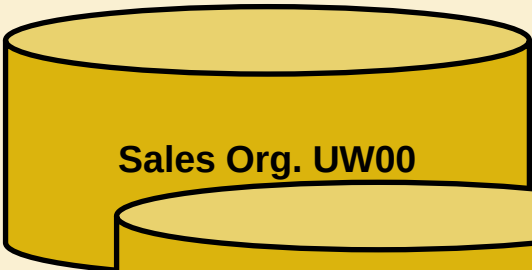
Material Master

General Information relevant for the entire organization:



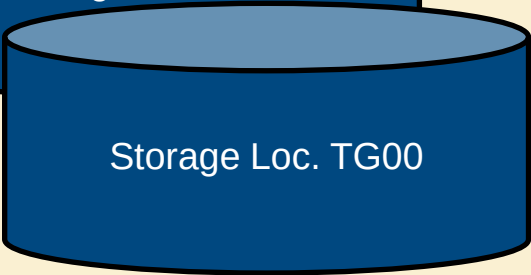
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

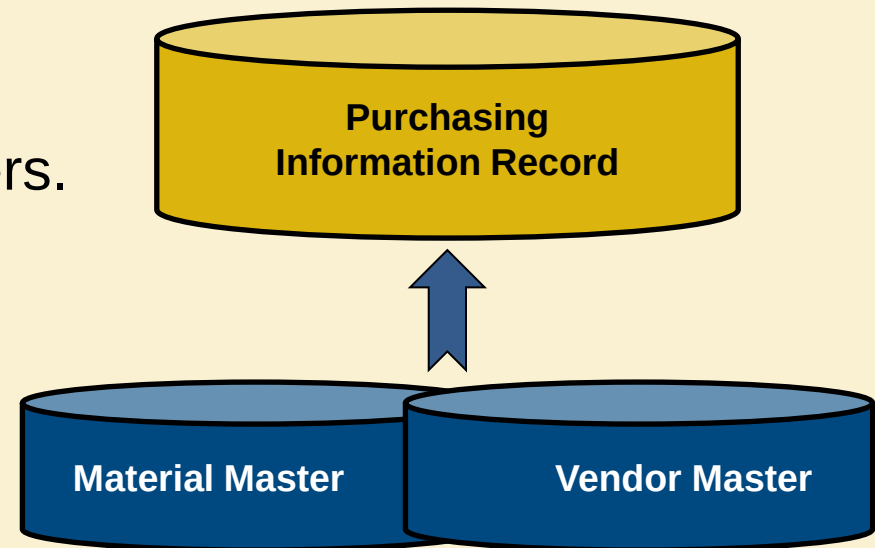
Storage Location specific information:



Stock Qty

Purchasing Information Record

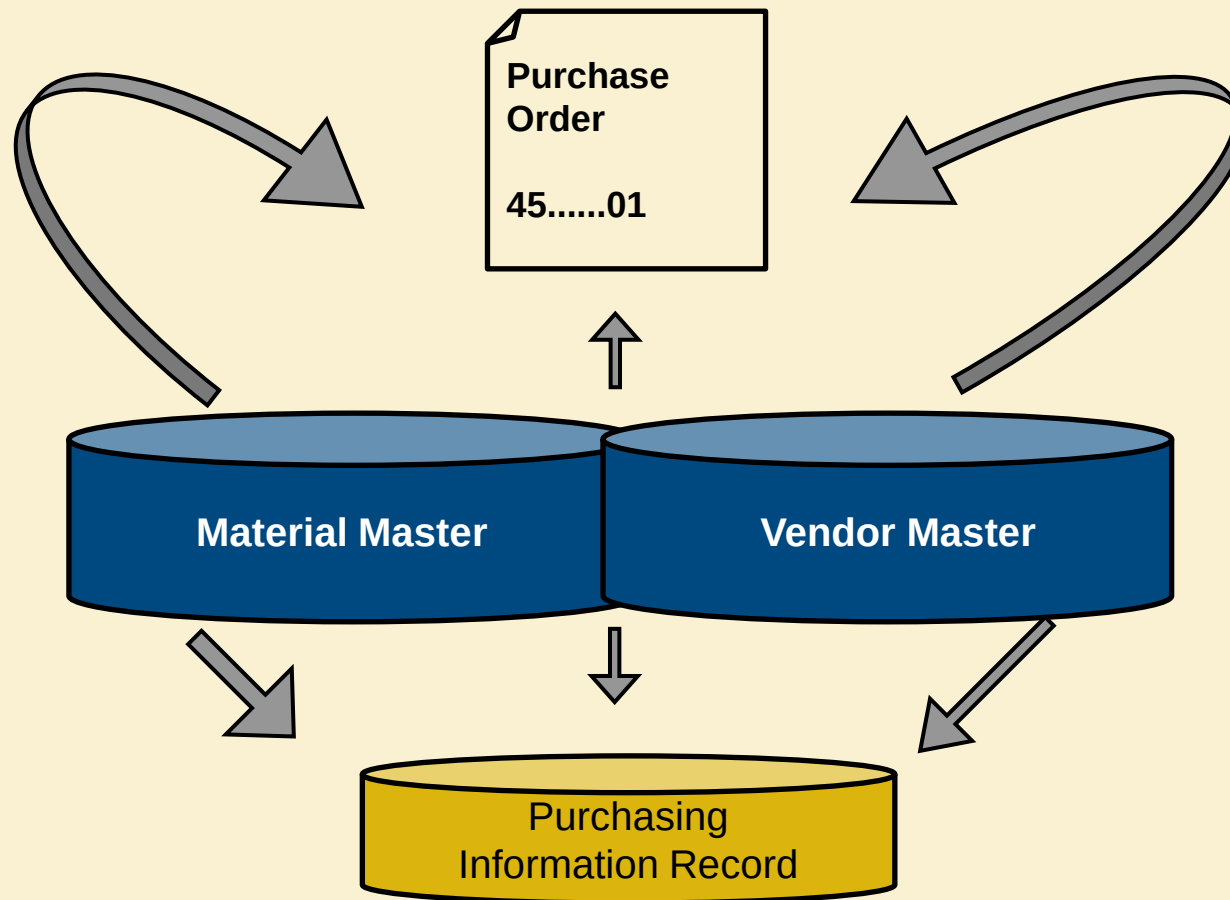
- Framework for Purchase Order
 - ❖ Contains the relationship between a vendor and a material
- Can be created:
 - ❖ Manually.
 - ❖ Automatically – Quotations.
 - ❖ Automatically – Purch. Orders.
- Reporting
 - ❖ Vendor Evaluation.



Purchasing Information Record (Cont.)

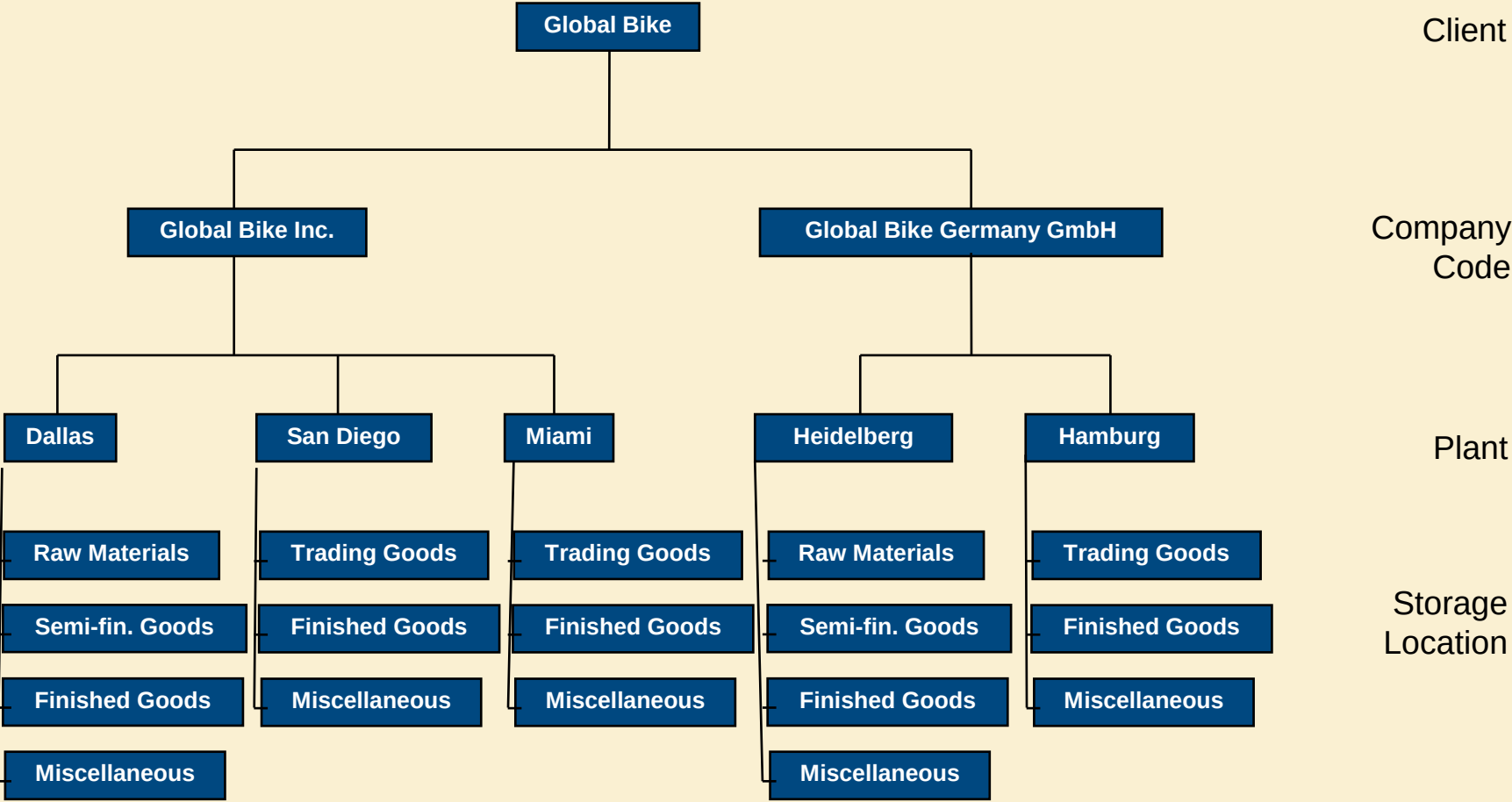
- Allows buyers to quickly determine:
 - ❖ Which vendors have offered or supplied specific materials.
- Info Records contain:
 - ❖ Data on pricing and conditions.
 - ❖ Last purchase order.
 - ❖ Tolerance limits for deliveries.
 - ❖ Specific lead times.
 - ❖ Availability periods.
 - ❖ Vendor Evaluation data.
- Serves as default information for Purchase Orders.

Master Data in Use



3. Procure-To-Pay Process

Global Bike Structure for Materials Management



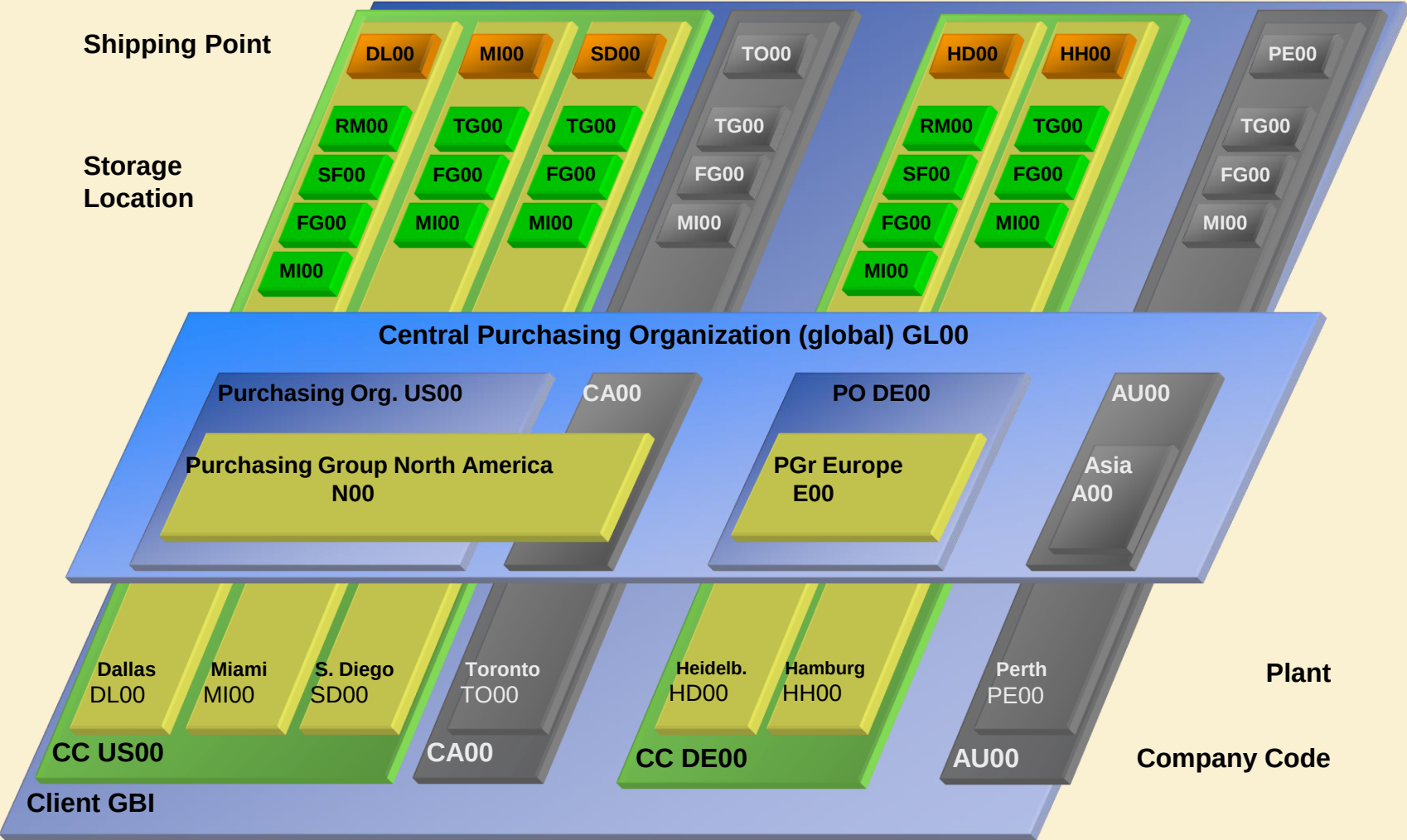
MM Organizational Structure (Materials Management)

- Client
 - ❖ An independent environment in the system.
- Company Code
 - ❖ Smallest organizational unit for which you can maintain a legal set of books.
- Plant
 - ❖ Operating area or branch within a company
 - Manufacturing, distribution, purchasing or maintenance facility.
- Storage Location
 - ❖ An organizational unit allowing differentiation between the various stocks of a material in a plant.

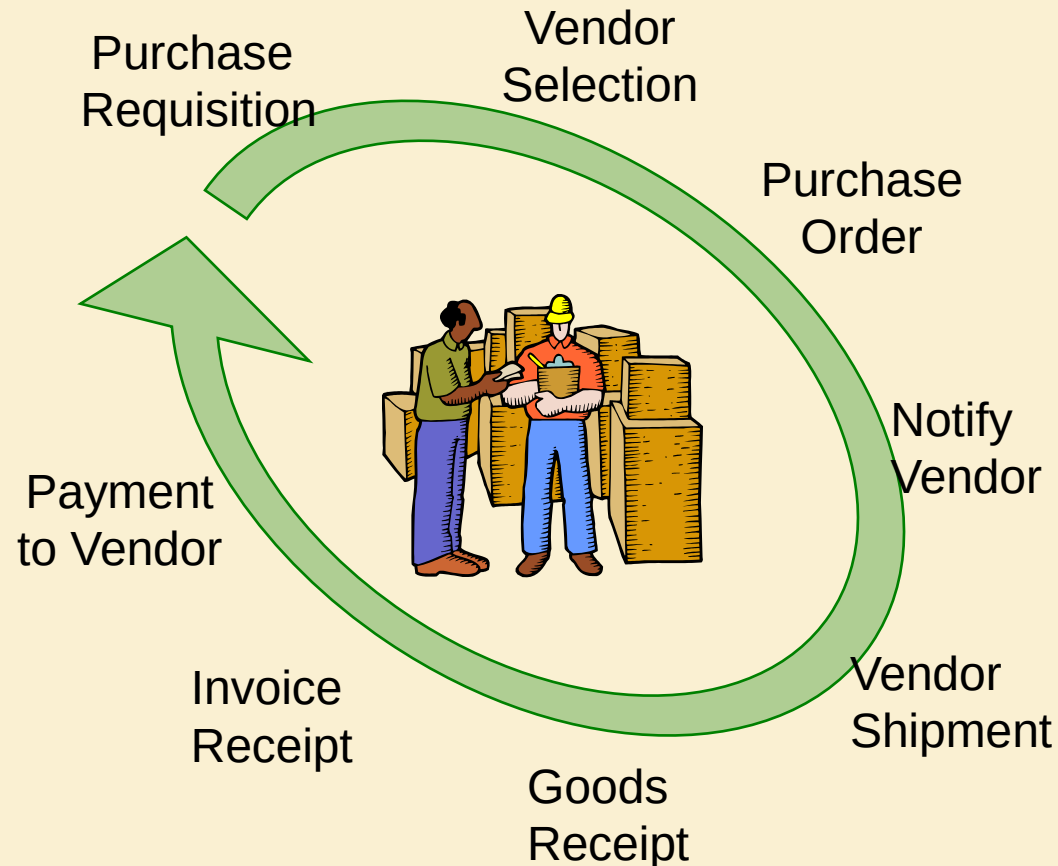
MM Organizational Structure (Purchasing)

- Purchasing Organization
 - ❖ The buying activity for a plant takes place at the purchasing organization.
 - ❖ Organization unit responsible for procuring services and materials.
 - ❖ Negotiates conditions of the purchase with the vendors.
- Purchasing Group
 - ❖ Key that represents the buyer or group of buyers who are responsible for certain purchasing activities.
 - ❖ Channel of communication for vendors.

Global Bike Enterprise Structure in SAP ERP (Logistics)



Procure-To-Pay Process



Purchase Requisition

- Internal Document instructing the purchasing department to request a specific good or service for a specified time
- Requisitions can be created two ways:
 - ❖ Directly - Manually
 - A person creating determines what, how much, and when.
 - ❖ Indirectly - Automatically
 - Materials Requirement planning (MRP).
 - Production Orders.
 - Maintenance Orders.
 - Sales Orders.

Requisition Sourcing

- Once the requisition has been assigned a source of supply it can be released for processing.
- There are a variety of ways that a purchasing department can process a requisition to determine the appropriate Source of Supply:
 - ❖ Internal Sourcing Requirements.
 - ❖ Source List.
 - ❖ Outlined Agreement.
 - ❖ Request for Quotation (RFQ).

Requisition Sourcing - Internal Sourcing

- The requisition for materials could be satisfied by sources within our company.
 - ❖ It is possible that a plant within your firm could represent a potential source of supply for the material needed (centralized warehouse).
 - ❖ If an internal source is identified the requirement is covered by an internal procurement transaction (stock transport order).

Requisition Sourcing - Source List

- A source list is a record that specifies the allowed means for procuring a material for a certain plant within a given time period.
 - ❖ If the list contains a sole source the system will assign the vendor to the requisition.
 - ❖ If several options exist, the system will display a list of vendors for you to choose from.
 - ❖ If no source has been established the system will revert to search information records and outline agreements.

Requisition Sourcing - Outline Agreement

- Requisitions can be satisfied through existing longer-term purchasing agreement.
- These agreements are subdivided into:

❖ Contracts

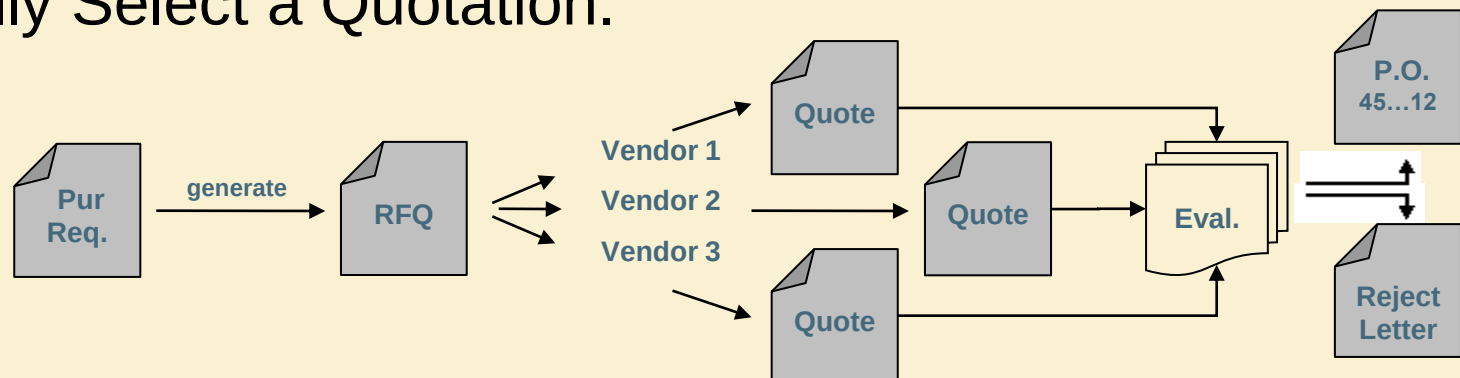
- Consists of items defining the individual materials, material groups, or services with prices and in many cases quantities.
 - Quantity.
 - Value.

❖ Scheduling Agreements

- Total quantity of material is spread over a certain period in a delivery schedule, consisting of line items indicating quantities and their planned delivery date.

Requisition Sourcing - Request for Quotation

- If nothing exist in the system, we may need to submit a request for quotation to our vendors. An RFQ is an invitation to a vendor by a Purchasing Organization to submit a bid for the supply of materials or services.
 - ❖ The accepted quotations will generate Purchasing Information Records.
 - ❖ Perform Quotation Price Comparisons.
 - ❖ Finally Select a Quotation.



Quotation from Vendor

- The quotation received by your company is a legally binding offer, should decide to do business with the vendor, containing prices and conditions for the materials specified in the RFQ for a predefined period of time.
 - ❖ In SAP, the RFQ and the Quotation will be become a single document, you will enter the vendor's response in the RFQ you created.

Vendor Evaluation once Identified

- Vendor evaluation helps purchasing evaluate vendors for sourcing while also enabling the company to monitor vendor relationships through performance scores and criteria you put in place.

- ❖ Supports a maximum of 99 main criteria and 20 sub criteria for each main:

Price

Price Level
Price History

Quality

Goods Receipt
Quality Audit
Complaints/Rejection level

Delivery

On-time delivery performance
Quantity reliability
Compliance with shipping instructions
Confirmation Date

- ❖ You then must establish a scoring range (1 -100) and determine the weight factors of scores for each.

Purchase Order

- A purchase order is a formal request to a vendor for a specific material or service under the stated conditions.
- Purchase Orders can be created manually.
 - ❖ Reference a Purchase Order.
 - ❖ Reference a Purchase Requisition.
 - ❖ Reference a RFQ/Quotation.
 - ❖ Without Reference.
- Purchase Orders can be created automatically.

Purchase Order (Cont.)

- A purchase order can be used for a variety of purposes, the item category (procurement type) defined in the PO will dictate the use of the order and the process that the order will follow:

The screenshot displays the SAP Purchase Order (PO) interface for document number 4500000000. The header section includes buttons for 'Edit', 'Copy', and 'Display Commitments'. Below the header, a table shows key details: Status is 'Follow-On Documents', Purchase Order Date is '02.08.2021', and Net Value is '5.250,00 USD'. The main content area is divided into two columns: 'Basic Data' and 'Organization'. The 'Basic Data' column lists 'Purchase Order Type: Standard PO (NB)', 'Supplier: Olympic Protective Gear (101999)', 'Currency: United States Dollar (USD)', and 'Language Key: English (EN)'. The 'Organization' column lists 'Purchasing Group: North America (N00)', 'Purchasing Organization: Global Bike US (US00)', and 'Company Code: Global Bike Inc. (US00)'. A 'Further Information' section at the bottom shows 'Created by: -', 'Purchase Order Date: 02.08.2021', 'Approval Status: -', and 'Approver: -'. The interface includes a navigation bar at the top with tabs for 'General Information', 'Items', 'Limit Items', 'Delivery and Invoice', 'Supplier Contact Data', 'Notes', 'Output Management', and 'Attachments'.

Purchase Order Details	
Status	Follow-On Documents
Purchase Order Date	02.08.2021
Net Value	5.250,00 USD

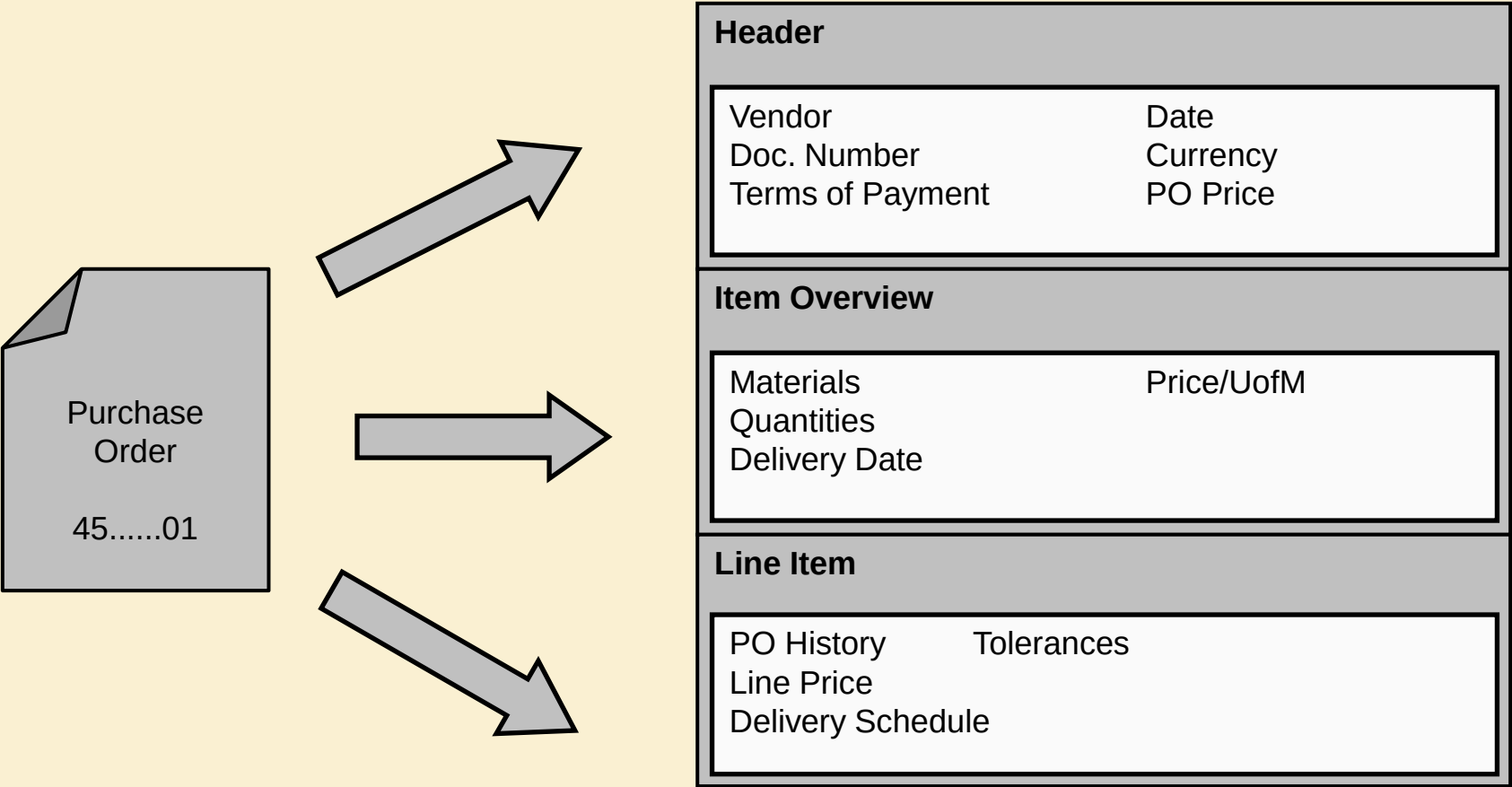
Basic Data	Organization
Purchase Order Type: Standard PO (NB)	Purchasing Group: North America (N00)
Supplier: Olympic Protective Gear (101999)	Purchasing Organization: Global Bike US (US00)
Currency: United States Dollar (USD)	Company Code: Global Bike Inc. (US00)
Language Key: English (EN)	

Further Information
Created by: -
Purchase Order Date: 02.08.2021
Approval Status: -
Approver: -

- ❖ Standard
 - Stock or Consumption
- ❖ Services
- ❖ Subcontracting
- ❖ Third-Party
- ❖ Consignment

Purchase Order (Cont.)

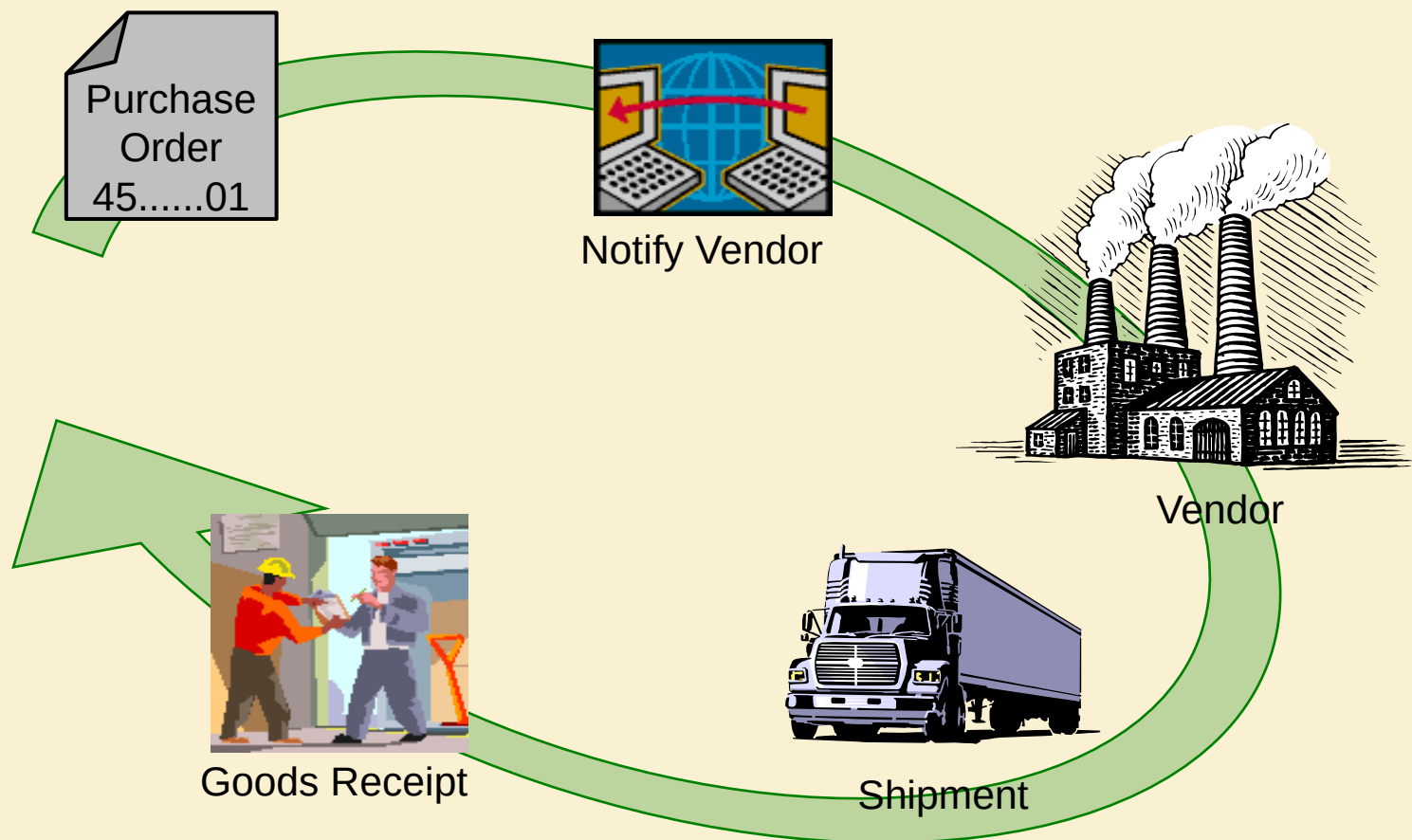
- Purchase Order Structure



Purchase Order (Cont.)

- Once a Purchase Order has been created the vendor needs to be notified
 - ❖ Printed; E-mail; EDI; Fax; XML.
- There are a variety of forms that aid in the purchasing process and are generated from the Purchase Order.
 - ❖ Purchase Order Output.
 - ❖ Order Acknowledgement Forms.
 - ❖ Reminders.
 - ❖ Schedule Agreements.

Goods Receipt



Goods Receipt (Cont.)

- Goods movement in which we accept goods into our system.
- If materials are delivered against a Purchase Order, we will reference that Order
 - ❖ Determine if we got what we ordered.
 - ❖ System can purpose data for us from the PO.
 - Material, quantity.
 - ❖ Purchase Order History is update with the receipt.
 - ❖ Updates Physical Inventory.
 - ❖ Updates Inventory G/L Account.

Material Movements

- When a goods movement takes place, it is represented by a Movement Type.
- Movement types are three-digit keys used to represent a movement of goods.
 - ❖ 101 – goods receipt into warehouse.
 - ❖ 103 – goods receipt into GR blocked stock.
 - ❖ 122 – return delivery to vendor.
 - ❖ 231 – consumption for a sales order.
 - ❖ 561 – initial entry of stock.
- Destinations for Receipt of Goods.
 - ❖ Warehouse – Unrestricted, Quality, Blocked.
 - ❖ Quality.
 - ❖ Goods Receipt Blocked Stock.

Effects of a Goods Receipt

- When a Goods Movement for the receipt of goods takes place a series of events occur
 - ❖ Material Document is Created.
 - ❖ Accounting Document is Created.
 - ❖ Stock Quantities are Updated.
 - ❖ Stock Values are Updated.
 - ❖ Purchase Order is Updated.
 - ❖ Output can be generated (GR slip / pallet label).

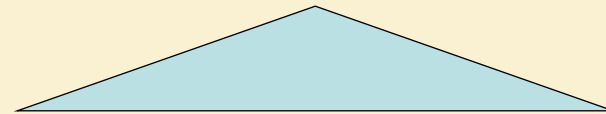
Invoice Processing

- Incoming Invoices are reference against a Purchase Order to verify their content, prices, and arithmetic.
- If differences arise between the purchase order or goods receipt and the invoice the system will generate a warning or an error.
 - ❖ Depending on system configuration the difference could cause the system to Block the Invoice.



Purchase order

- Target quantity
- Target price



Invoice receipt

- Actual price



Goods receipt

- Actual quantity

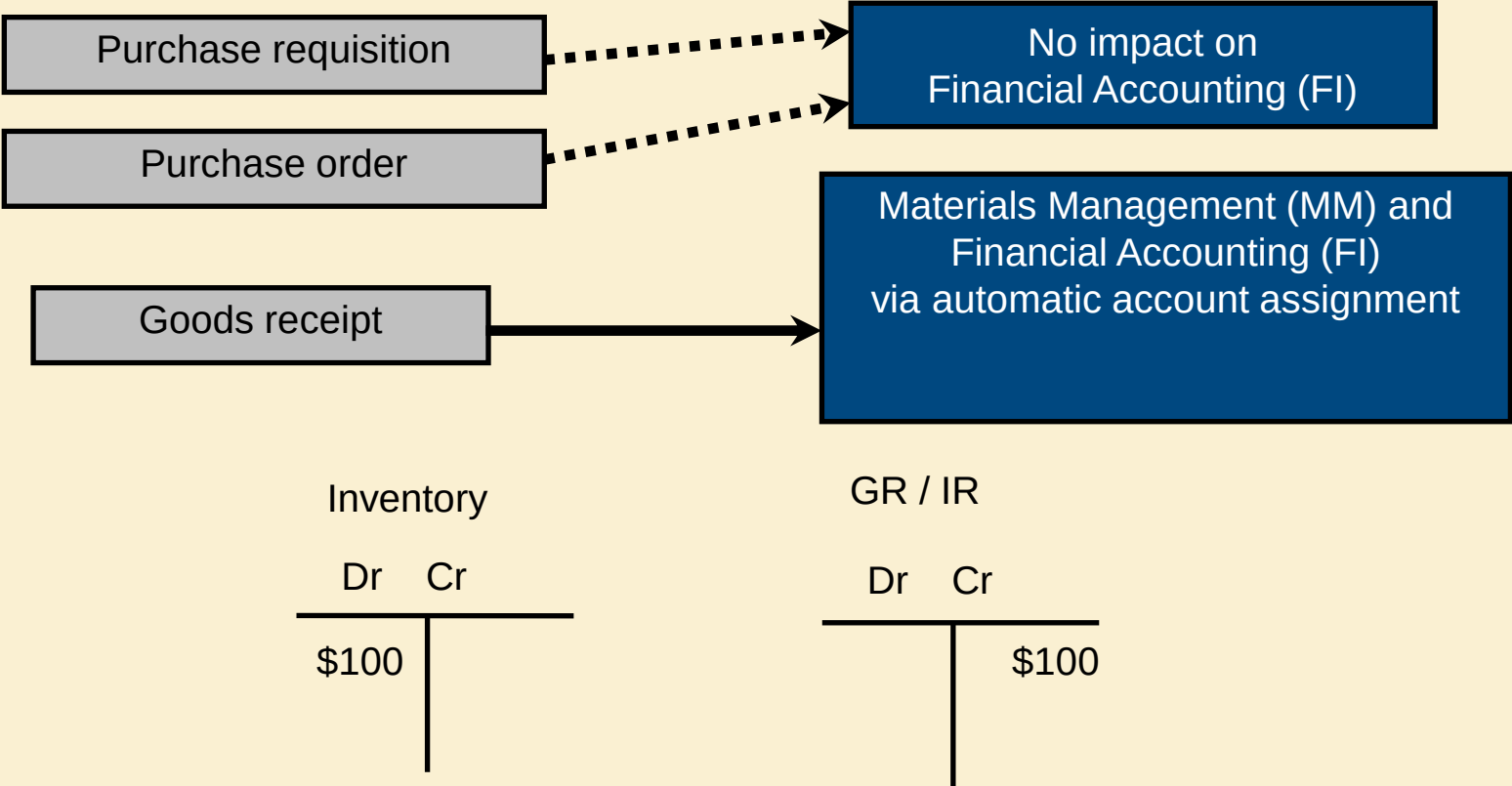
Invoice Processing (Cont.)

- When an invoice is saved, it applies the liability from the Goods Receipt of our Purchase Order to a Vendor.
- Upon verification the:
 - ❖ Purchase Order is updated.
 - ❖ Material Master is Updated (MAP).
 - ❖ Accounting Document is created.
- Once the Invoice has been posted, the verification process is completed, and the payment process is initiated within Financial Accounting (FI).

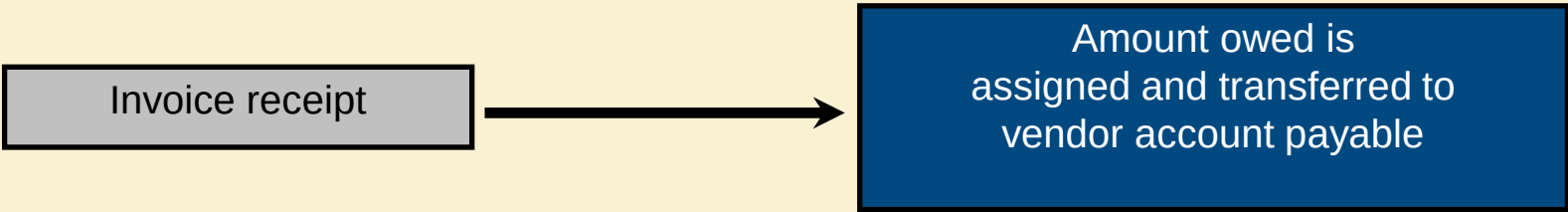
Payment to Vendor

- Can be done automatically or manually.
 - ❖ Post Outgoing Payment vs. Payment Program.
- Elements of the Payment Transaction:
 - ❖ Payment Method.
 - ❖ Bank from which they get paid.
 - ❖ Items to be paid.
 - ❖ Calculate payment amount.
 - ❖ Print payment medium.
- Process will create a financial accounting document to record the transaction.

Goods Receipt / Invoice Receipt Reconciliation Account

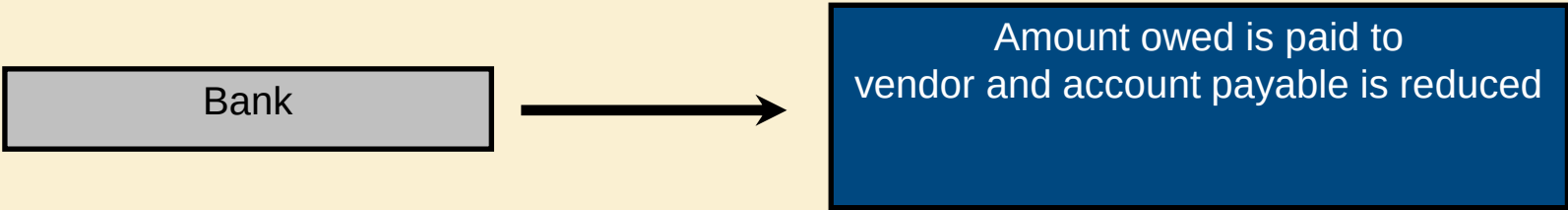


Goods Receipt / Invoice Receipt Reconciliation Account (Cont.)



GR / IR		Vendor A/P	
Dr	Cr	Dr	Cr
\$100			\$100

Vendor Payment



Bank		Vendor A/P	
Dr	Cr	Dr	Cr
	\$100	\$100	

FI – MM Integration Point

Goods
Receipt

Invoice
Receipt

Payment
Program

Inventory		GR / IR		AP (Vendor)		Bank	
Dr	Cr	Dr	Cr	Dr	Cr	Dr	Cr
\$100		\$100	\$100	\$100	\$100		\$100

In conclusion

- Procure-To-Pay Process.
- Purchase Requisition.
- Vendor Selection.
- Purchase Order.
- Notify Vendor.
- Vendor Shipment.
- Goods Receipt.
- Invoice Receipt.
- Payment Vendor.
- ...

Understand



4. Materials Management in SAP S/4HANA (using Global Bike)

**THANK YOU
FOR YOUR ATTENTION**

Q&A